WhisperWard OSINT: Case Study

A child-safety OSINT investigation platform with explainable, human-in-the-loop risk scoring.

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Live demo: whisperward-osint.onrender.com

Overview

WhisperWard is a deployed, open-source intelligence (OSINT) platform designed to support investigations into online child exploitation on gaming and social platforms.

I designed and built the entire system end-to-end. It guides investigators from initial subject discovery through to court-defensible evidence packaging by combining automated public data collection, cross-platform identity correlation, a fully transparent weighted risk-scoring engine, and forensic evidence handling, all backed by mandatory human review at its core.

This is not a prototype. WhisperWard runs a live public demonstration environment, carries 411 automated tests in continuous integration, and is published under AGPL-3.0 so its safeguards remain open and auditable.

WhisperWard boot screen — operator shown as unauthenticated until sign-in, with the live active-case count

The Problem

Online child exploitation on platforms like Roblox and Discord continues to grow. These environments are primary spaces for children to socialize, and predators heavily target them for grooming.

Investigators in law enforcement, platform Trust & Safety, and nonprofits face significant challenges: grooming signals are fragmented across platforms and time, and existing tools often lack the ability to correlate data and produce defensible, explainable assessments.

Two core constraints shaped WhisperWard's architecture:

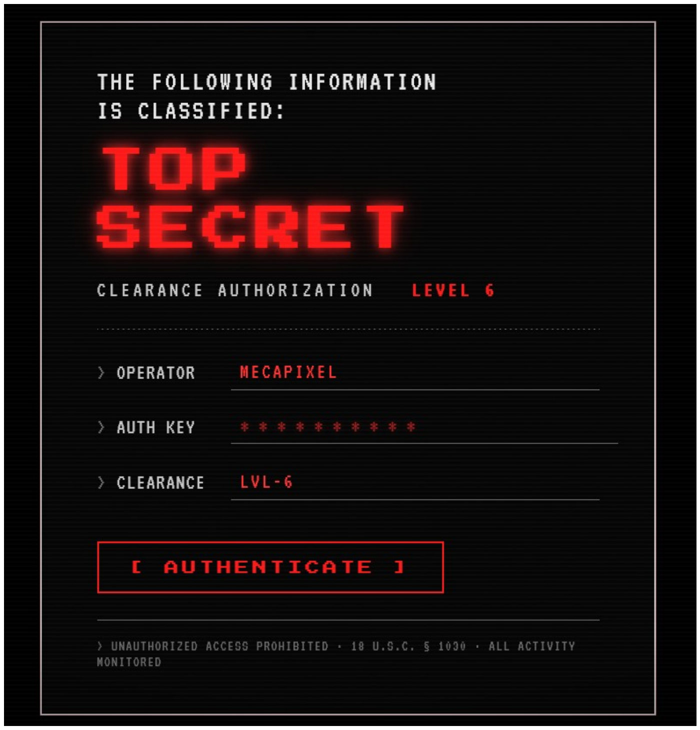
Fragmented, cumulative signals. Grooming is rarely evident in a single message. It emerges as patterns such as age probing, boundary testing, secrecy requests, and platform migration, unfolding across time and accounts. The tooling must reason about the pattern, not the message.

Explainable, defensible assessments. Any system that may inform referrals or legal processes must let reviewers trace every risk point to its source. A black-box score is unacceptable in this domain.

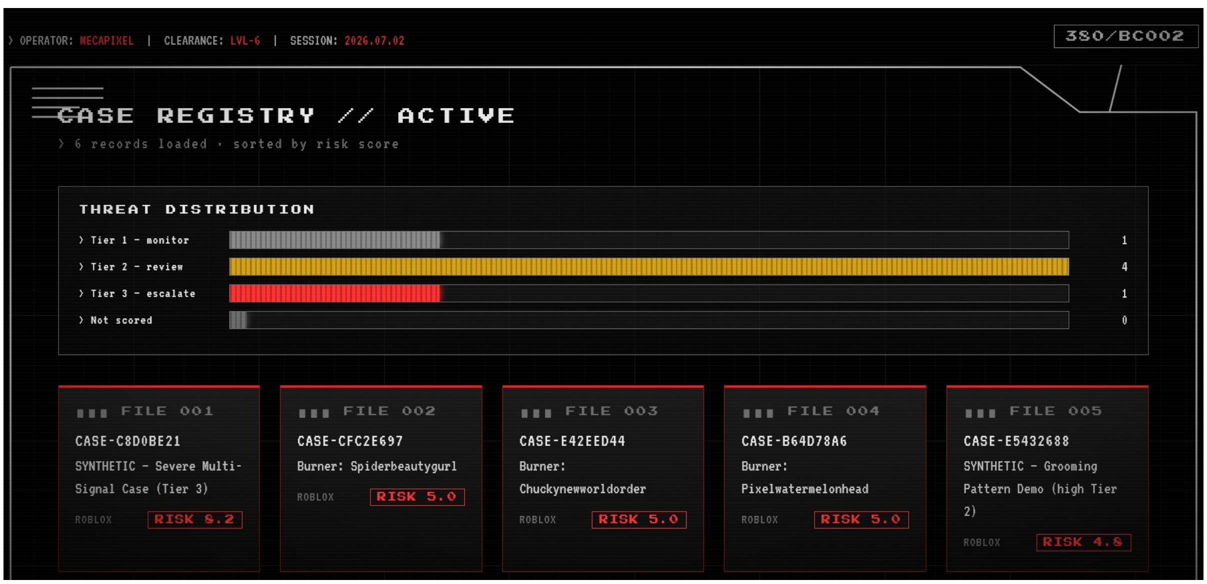
What WhisperWard Does

The platform executes a complete investigative pipeline:

1. **Collection.** Gathers public profile data from platforms like Roblox and correlates usernames across sites.
2. **Correlation.** Links identities using fuzzy matching, writing-style analysis, and perceptual avatar hashing, exposed through a first-class CLI correlate command.
3. **Behavioral analysis.** Detects grooming patterns in chat content, accounting for multi-step sequences.
4. **Risk scoring.** Generates a weighted, fully explainable risk score with tier recommendations.
5. **Evidence & reporting.** Packages artifacts with SHA-256 hashing and chain-of-custody manifests, and supports redaction and referral export.
6. **Human review.** Every escalation requires human approval. The system provides evidence and reasoning. It never makes autonomous decisions.



Operator authentication — a signed session is required before any case data is shown



Case registry — active cases sorted by risk, with the D3 threat-tier distribution

Full registry — six active cases: two synthetic severe demos plus four investigator-created burner accounts

Classified case file — quick-view popup

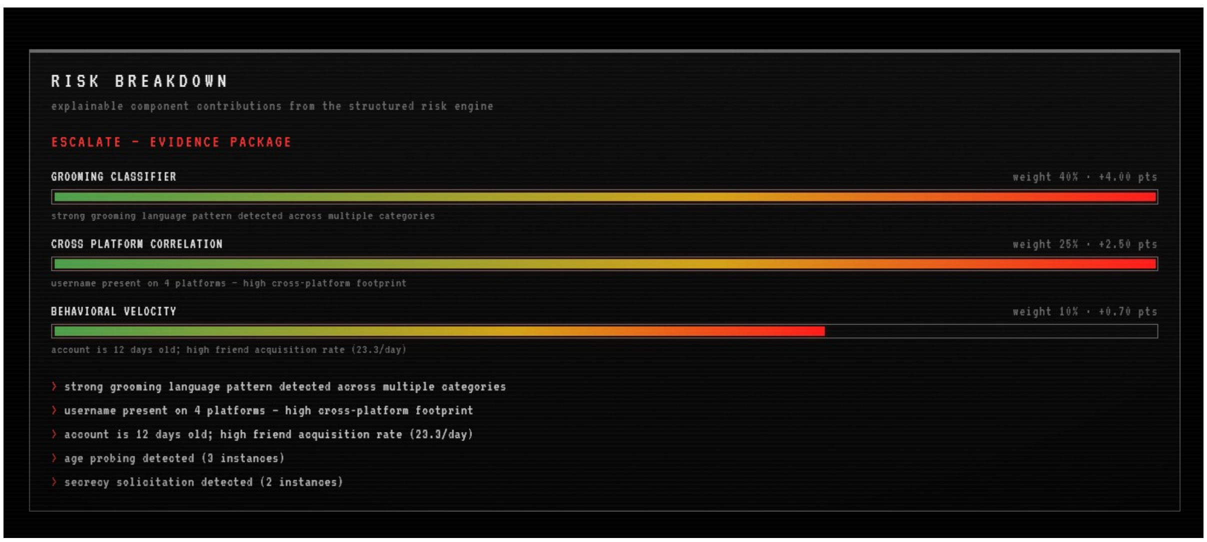
The Explainable Risk Engine (Core Differentiator)

The risk engine is the heart of WhisperWard. Instead of opaque scores, it evaluates subjects across five weighted components, each with traceable contributions and plain-language explanations:

Component	Weight	What it measures
Grooming language	40%	Multi-category pattern detection in chat content
Cross-platform correlation	25%	Same identity detected across multiple platforms
Anonymization	15%	Use of Tor / VPN infrastructure
Behavioral velocity	10%	Account age and rapid friend acquisition
Historical signals	10%	Prior flags in the investigation database

A synergy bonus rewards corroborating signals. Scores map to three tiers: **Tier 1 (Monitor)**, **Tier 2 (Human Review Required)**, and **Tier 3 (Escalate)**. The Tier 3 threshold is deliberately set above what any single-platform signal can reach on its own, so escalation requires genuine cross-platform or historical corroboration.

The result: every score is fully accountable. A reviewer sees not “8.2,” but “8.2 because grooming language contributed 4.0 points, cross-platform presence 2.5, behavioral velocity 0.7,” each with its own plain-language explanation.



Risk breakdown panel showing weighted components and the terminal intel log

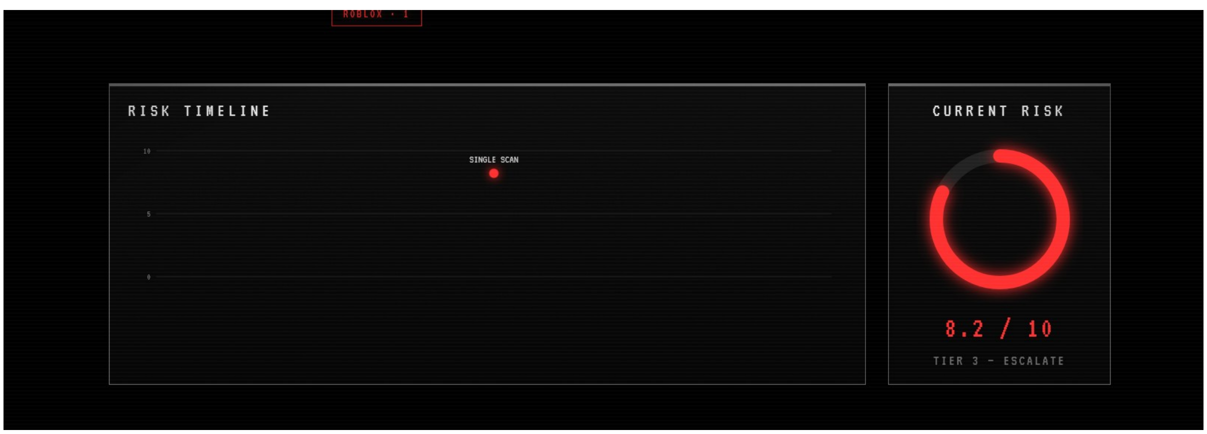
Demonstration

The live demo carries six active cases. All use investigator-created test accounts and fabricated data, which the project’s governance framework permits — no real minor’s data and no third-party accounts are processed. Four are burner accounts created for testing; the two synthetic demo cases below were built from that same fabricated material to exercise the engine across its full range and show that it discriminates severity based on real signals rather than flagging everything alike.

Tier 2, Human Review Required (score 4.8). A synthetic subject exhibiting a grooming-language pattern (age probing, a compliment, platform-migration pressure, and a secrecy request) across two platforms on a young account. The engine flags it for review without over-escalating, which is the realistic, common case.

Tier 3, Escalate (score 8.2). A synthetic subject exhibiting a strong multi-category grooming pattern across four platforms on a very new, high-velocity account. Every axis fires, the corroboration synergy applies, and the engine escalates to an evidence package with mandatory human sign-off.

Synthetic case dossier (Tier 3, score 8.2) with explainable scoring



Risk timeline and current-risk gauge

The contrast between the two demonstrates the design goal: the score is not a gut feeling, it is a defensible sum of documented parts, and the same engine correctly separates a review-level case from an escalation-level one.

Governance, Ethics, and Safety

Safeguards are foundational, not afterthoughts:

Human-in-the-loop by design. No autonomous decisions. Every escalation requires human sign-off before any action.

Approval-gated CSAM handling. Perceptual-hash detection is disabled by default and gated behind explicit approval and credentials available only to vetted organizations. It never stores image content, processes hashes in memory only, and flags every match for mandatory human review.

Synthetic data only. All demonstration and testing uses clearly labeled fabricated data and investigator-created test accounts. No real minor's data and no third-party accounts are processed.

Enforced operator identity. The web dashboard requires an authenticated operator session, signed and verified server-side, before any case page is displayed, and the authoritative operator record lives in the tamper-evident chain-of-custody log.

Open and auditable. Published under AGPL-3.0 specifically so the safeguards cannot be quietly stripped out.

Engineering

Stack. Python, FastAPI, SQLite, D3.js, with a modular, plugin-based collection architecture.

Quality. 411 automated tests running in GitHub Actions on every push. The scoring engine, behavioral classifier, and correlation modules are independently tested, and the suite runs clean from a fresh clone in under twenty seconds.

Architecture. A clean separation between collection, correlation, scoring, and evidence layers; an explainable scoring engine decoupled from data collection so scores are reproducible and traceable; and a web layer that reads the same persisted data the command-line tools produce, so the dashboard and the reports always tell the same story.

Interface. A full command-line tool for investigators plus a session-gated web dashboard for review and visualization, including the risk-timeline, tier gauge, and explainable breakdown views.

What's Next

Model-assisted behavioral analysis. Evaluating small, fine-tunable language models as an additional signal feeding the structured engine, kept as an explainable component rather than an opaque score.

Deeper platform coverage. Extending collection and correlation to additional platforms central to how minors socialize online, starting with a Discord module built on the existing plugin contract.

Collaboration. Engaging with child-safety organizations and platform Trust & Safety teams that build and use tooling in this space.

Summary

WhisperWard demonstrates the ability to take a hard, sensitive problem and build a complete, responsible, production-grade solution: robust data collection, transparent risk assessment, forensic evidence handling, and strong ethical guardrails. It is public, deployed, thoroughly tested, and built to the standard this work demands.

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